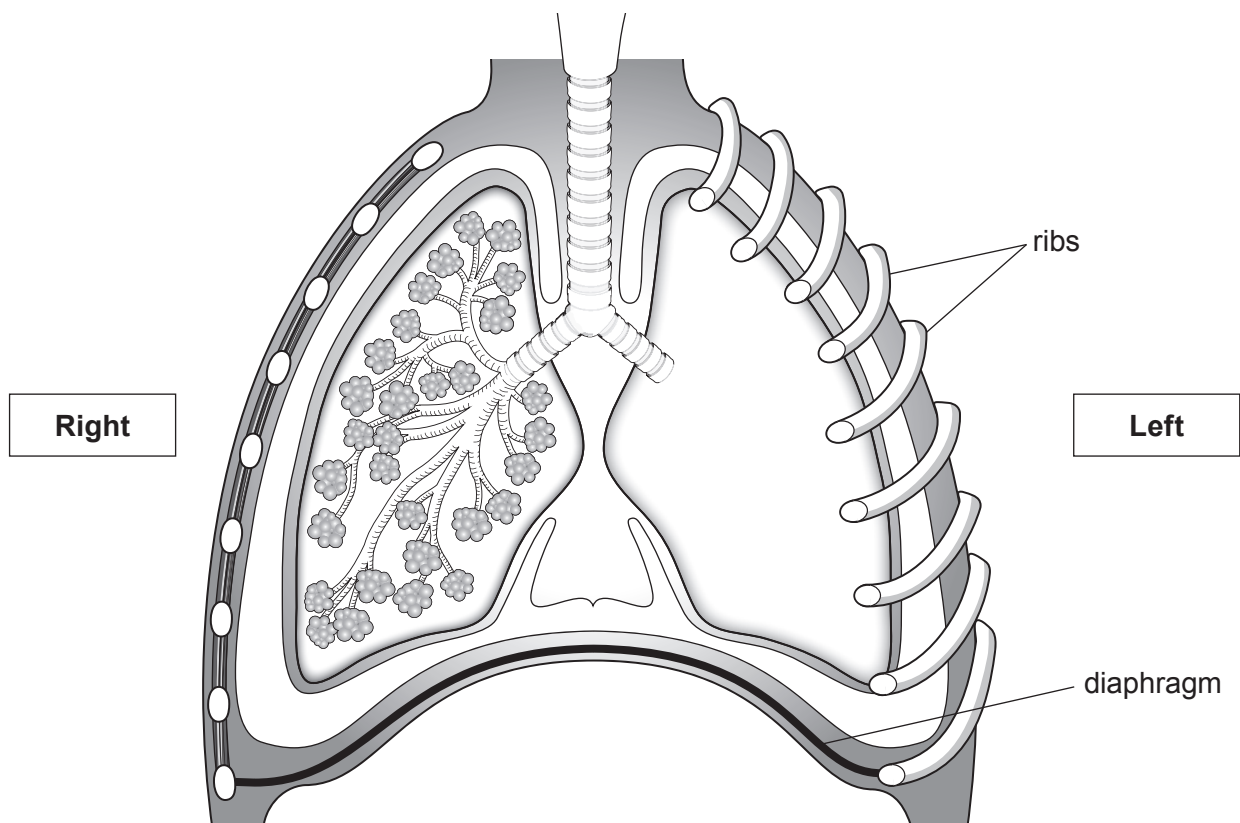


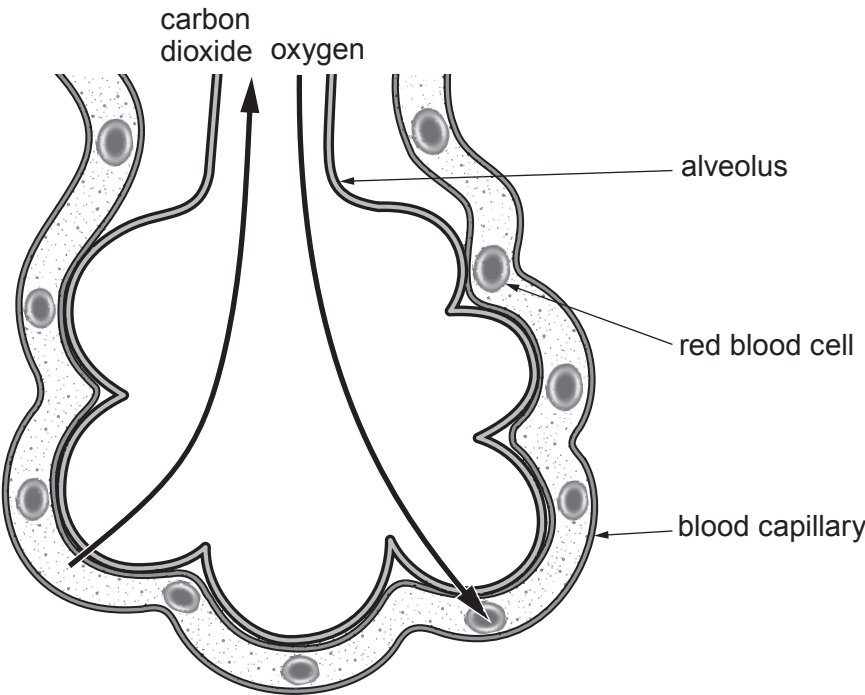
4. (a) The diagram below is a section through the human thorax during expiration (breathing out).



On the diagram, **draw**

- (i) **a line** to show the shape of the diaphragm at the end of inspiration (breathing in); [1]
- (ii) **an arrow** on the diagram to show the path taken by air passing from the trachea to a bronchiole in the **right** lung. [1]

(b) The diagram below shows a single alveolus.



Gas exchange between the alveolus and the blood in the capillary occurs through the wall of the alveolus.

Answer the following questions using information from the alveolus diagram above and your own knowledge.

(i) Name the process by which gas exchange occurs. [1]

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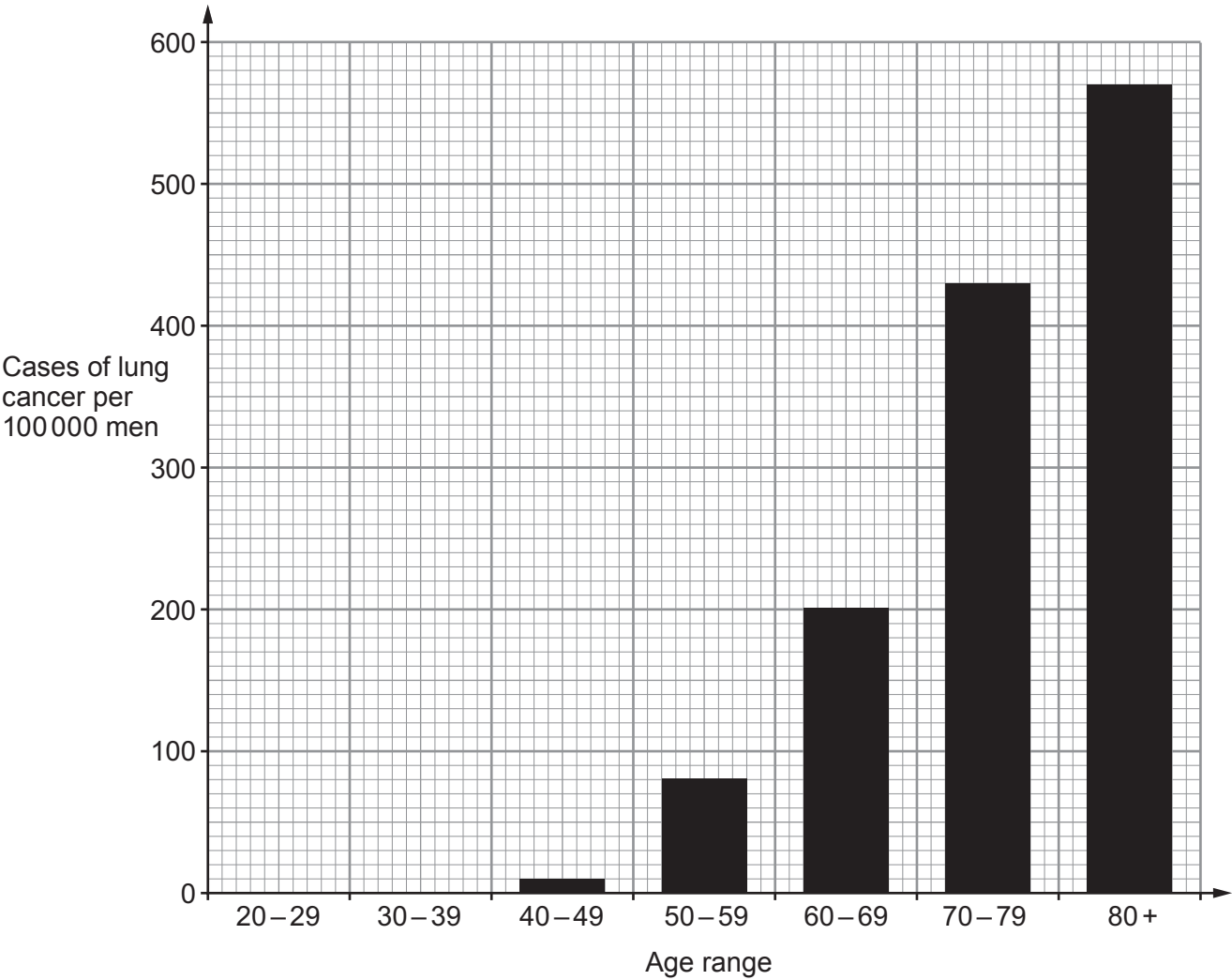
(ii) Describe **two** ways in which the structure of the alveolus is adapted for this function. [2]

1.

2.

Examiner
only

- (c) Cigarette smoking damages the lungs and causes lung cancer. In 2009, doctors investigated lung cancer and cigarette smoking in men in a city in the UK. The highest proportion of smokers was found to be in the age range 20-29. The cases of lung cancer are shown in the graph below.



- (i) State the conclusion the doctors should reach using the data given in the graph. [1]

- (ii) What is the evidence that lung cancer takes a long time to develop? [1]

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